7(a)	mean power $\langle P \rangle = I_{r.m.s.}^2 R$ $= \left(\frac{2.7}{\sqrt{2}}\right)^2 (50)$ $= 182.3$ W ≈ 180 W (2 s.f.)	[1] for substitution [1] for correct mean power
7(b)	r.m.s. voltage across secondary coil $V_S = I_{r.m.s.} R$ $= \left(\frac{2.7}{\sqrt{2}}\right)(50)$ $= 95.46$ V turns ratio $\frac{N_P}{N_S} = \frac{V_P}{V_S}$ $\frac{25}{N_S} = \frac{20}{95.46}$ $N_S = \frac{(25)(95.46)}{20}$ $= 119$ ≈ 120 (2 s.f.)	[1] for correct V_s [1] for correct turns ratio [1] for correct N_s
7(c)	For the current in the secondary coil, period $T = \frac{50 \times 10^{-3}}{3} = 16.7 \times 10^{-3}$ s frequency $f = \frac{1}{16.7 \times 10^{-3}}$ $= 60$ Hz Hence, the frequency of the alternating voltage supply is 60 Hz, since it is the same as the frequency of the current through the secondary coil.	[1] for correct substitution [1] for correct frequency
7(d)	The values of the root-mean-square current and voltage of the alternating voltage supply are independent of its frequency. Hence, the mean power due to the alternating voltage supply and hence the mean power dissipated across R remains unchanged.	[1] [1]
8(a)(i)	Newton's law of gravitation states that the force of attraction	[2]